Nishi Baranwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.24
PACE JUNIOR SCIENCE COLLEGE, ANDHERI	Class XII (MAHARASHTRA STATE BOARD OF SECONDARY & HIGHER SECONDARY EDUCATION)	2022	93.50%
HIRANANDANI FOUNDATION SCHOOL, POWAI	Class X (COUNCIL FOR THE INDIAN SCHOOL CERTIFICATE EXAMINATIONS)	2020	96.80%

EXPERIENCE

Software Engineering Internship, Databricks Bengaluru

- Automated TiDB/MySQL (across 3 clouds) disaster recovery system in **Scala** & **Temporal**, making it **75%** faster.
- Reduced **55+** step SOPs to a single command, ensuring reliability via 10+ 2PA safety gates with resumability.
- Built a **Protobuf/gRPC** server with **Kafka**-backed state in the CDC-Ingester for dynamic reconfiguration.
- Developed **Scala** workflows for Cloud DB recovery, permission revocation, and automated instance rename-swaps.

PROJECTS

Intelligent Event Management | Guide: **Prof. Manish Singh**

- Built a **multi-tenant SaaS** platform using **React, Node.js & MongoDB** via microfrontend architecture in a team of 6.
- Implemented a dynamic template engine, RBAC, structured event lifecycles, QR check-ins & real-time registrations.

Concurrent Merkle Tree Operations | Research | Guide: **Prof. Sathya Peri**

- Built a **linearizable** concurrent Merkle tree update algorithm via fine-grained, hand-over-hand locking in a team of 2.
- Implemented a branch snapshot algorithm and lock-free reads while maintaining strict linearizability.
- Evaluated against sequential and Angela implementations, achieving higher throughput and lower response times.

Image Classification System | Guide: **Prof. Srijith P.K.**

- Built a 14.3M-param **SE-ResNet** in **PyTorch** with MixUp & OneCycleLR to classify 3D shapes in a team of 3.
- Extracted 272 **HOG/LBP** features to train an **XGBoost** classifier, mitigating shadow artifacts.
- Combined CNN & **XGBoost** via 8-pass TTA & dynamic thresholding, achieving **75.4%** test accuracy.

TransactiWar (Secure App) | Guide: **Prof. Saurabh Kumar**

- Built a secure **PHP/SQL** transaction app in a team of 5 with **Argon2id** hashing, ACID transfers, & secure image uploads.
- Integrated defenses against **XSS**, **SQLi**, and **CSRF** using PDO statements, rotating tokens, and rate limits.
- Exploited **PHP** array injections and client-state flaws in opponent apps to bypass logic & hijack sessions.

Finex DSL | Guide: **Prof. Ramakrishna Upadrasta**

- Built a finance DSL in **C++** via **Flex/Bison** and a memory-safe AST using **std::unique_ptr** & Visitor Pattern.

- Developed a 2-pass semantic analyzer and nested symbol table to resolve types, scope, and forward declarations.

Intent-Based Networking | Guide: **Prof. Kotaro Kataoka & Prof. Praveen Tammana**

- Built a **Mininet**-emulated SDN IBN system in a team of 6 using **RASA** NLU to translate intents into network policies.
- Automated **P4** rule deployment on **BMv2** switches via **gRPC** for connectivity & firewall policies.
- Implemented **Dijkstra** routing with dynamic self-healing to automatically reroute upon link drops.

LLVM IR Open Source Contribution | Guide: **Prof. Ramakrishna Upadrasta**

- Implemented a **C++** FP-based ctlz expansion for LLVM X86 SSE2 targets, cutting cycles by **~53%**.
- Ported an NVPTX rem custom combine to the generic DAGCombiner to enable global CSE optimizations.

Dashboard Admin Console | IITH Lambda Club | Core Team Member

- Collaborated on the IITH Dashboard website and app providing admin services.
- Developed app frontend in **Dart** using **Flutter**, integrating **Firebase** auth & **Dio** APIs to manage user data.
- Built the faculty room-booking system web frontend in **React + ShadCN**.

RISC V (64I) Simulator | Guide: **Prof. Rajesh Kedia**

- Implemented a **C++** RISC-V simulator featuring a custom assembler, a CPU executor and compile/runtime debugger.
- Simulated memory, cache & registers via **C++ STL maps & vectors** for efficient execution.

SKILLS

- **LANGUAGES:** C++, C, Python, Dart, JavaScript, Go, SQL, Scala, PHP, P4
- **SOFTWARE/TOOLS:** React.js, Node.js, Git, Flutter, Google Apps Script, Mininet, MongoDB, Pytorch, Wireshark, gRPC
- **OPERATING SYSTEM:** Ubuntu (Linux - based), MacOS (Unix - based)

RELEVANT COURSES

Software Engineering, Compilers, Computer Networks, Deep Learning, Operating Systems, Database Management Systems, Network Security, Parallel & Concurrent Programming, Introduction to Blockchain Programming, Computer Architecture

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for **DBMS-I & OS-I** under the guidance of Prof. Manish Singh and Prof. Sandipan Dandapat.
- Core Team Member at **Lambda**, The Software Development Club of Sci-Tech Council, IIT Hyderabad 2024-2025.
- Deputy Coordinator for Tech Events at **Elan & Nvision 2024**, the Cultural and Technical Fest of IIT Hyderabad.

SCHOLASTIC ACHIEVEMENTS

- Selected for **INSPIRE** Scholarship for securing **Top 1%** in M.H.S.C. 2022.
- Secured **31st rank** in **All India Open Mathematics Scholarship Examination** 2019 by I.P.M.